

PROJECT REPORT

IMPORT DATA USING TRANSFORM MAPS

ServiceNow Data Import and Transformation Project

Project Name	Import Data using Transform Maps
Project Design	Phase-II
Report Type	Final Project Report
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Date	01 October 2026

Prepared using the supplied Project Report Format and the project-specific Problem–Solution Fit, Agile Planning, UAT, and Customer Journey references.

1. INTRODUCTION

1.1 Project Overview

Import Data using Transform Maps is a ServiceNow-oriented data ingestion and transformation project. The project addresses the controlled movement of legacy CSV/Excel data into ServiceNow tables using Import Sets and Transform Maps. The project focuses on clean data migration, correct source-to-target mapping, duplicate prevention, data validation, and maintenance of referential integrity.

1.2 Purpose

The purpose of the project is to provide a structured approach for bulk data onboarding into ServiceNow. The proposed solution uses Coalesce keys for deduplication, validation scripts for data quality, reference-field handling for related records, and role-based access controls for secure processing.

2. IDEATION PHASE

2.1 Problem Statement

Organizations importing employee, asset, or organizational data into ServiceNow can face duplicate records, inconsistent formatting, incorrect field mappings, broken reference fields, and time-consuming manual data cleansing. The project proposes a controlled Import Set and Transform Map workflow to reduce these data-import problems.

2.2 Empathy Map Canvas

Says / Thinks	Does	Feels	Needs
"I need the data imported correctly." "I do not want duplicate records." "Reference fields must remain correct."	Cross-references spreadsheets. Cleans source data. Runs imports and fixes broken records.	Before: frustrated, anxious, overwhelmed. After: confident, relieved, in control.	Reliable field mapping. Validation. Deduplication. Secure bulk onboarding.

2.3 Brainstorming

- Use Import Sets as the controlled staging point for source data.
- Use Transform Maps to map source fields to target fields.
- Use Coalesce keys such as the employee identifier to reduce duplicate records.
- Use validation logic before records are committed to the target table.
- Use reference-field resolution for related users and departments.
- Use ACLs and role-based access to support secure data onboarding.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The customer journey follows Entice, Enter, Engage, Exit, and Extend. For this project, the journey begins when an administrator identifies a bulk-import requirement and continues through preparation, transformation, verification, and reuse of the import process.

Stage	Steps	Interactions / Touchpoints	Goal / Motivation
Entice	Identify bulk data import need	ServiceNow documentation; internal data-governance discussion	Understand the need for a controlled import process
Enter	Prepare CSV/Excel and Import Set	Source file; Import Set table; source fields	Prepare data correctly and avoid manual errors
Engage	Map, transform, validate and deduplicate	Transform Map; field mappings; Coalesce; validation; reference fields	Import accurate records without duplicates or broken references

Stage	Steps	Interactions / Touchpoints	Goal / Motivation
Exit	Verify transformed target records	Target table; transform history; logs	Confirm that imported records are correct
Extend	Reuse and maintain the process	Saved Transform Map; validation rules; ACLs	Make future data loads more consistent and controlled

3.2 Solution Requirement

- A source-data staging mechanism using Import Sets.
- A Transform Map for source-to-target field mapping.
- A Coalesce key for duplicate detection/update behaviour.
- Validation logic for missing, invalid, or inconsistent source values.
- Reference-field resolution for related ServiceNow records.
- Transform logging and verification of target records.
- Role-based access controls for secure data onboarding.

3.3 Data Flow Diagram

SOURCE	STAGING	TRANSFORMATION	TARGET	VERIFICATION
CSV / Excel Employee / Asset / Organizational Data	Import Set Import Set Table	Transform Map Field Mapping Coalesce Validation	ServiceNow Target Table	Transform History Logs Record Verification

Data flows from the source file into an Import Set, where the data is staged. The Transform Map then maps and transforms source values before loading them into the target ServiceNow table. The final stage verifies records and checks transformation results.

3.4 Technology Stack

Technology / Component	Purpose
ServiceNow	Platform for Import Sets, Transform Maps, target tables and access control
Import Sets	Temporary staging area for imported source data
Transform Maps	Maps and transforms source data into target records
ServiceNow JavaScript	Validation / transformation scripting
Coalesce	Supports duplicate detection and record matching
Reference Fields / Dot-walking	Resolves related records such as users and departments
ACLs	Role-based access control for secure data handling
CSV / Excel	Source data format for bulk import

4. PROJECT DESIGN

4.1 Problem Solution Fit

Dimension	Project-specific analysis
Customer Segment	Enterprise ServiceNow administrators, IT service desk managers, HR data migration specialists, and system integrators.
Jobs-to-be-Done	Migrate legacy CSV/Excel data, prevent duplicates, map source columns correctly, and maintain reference integrity.

Dimension	Project-specific analysis
Triggers	Data corruption, duplicate records, organizational restructuring, and broken reference-field errors.
Available Solutions	Manual CSV imports, external ETL scripts, and basic Transform Maps without validation scripts.
Root Cause	Lack of standardized validation, contextual relational mapping, intelligent deduplication, and manual-entry errors.
Solution	Import Sets and Custom Transform Maps with Coalesce, validation, reference resolution, and ACLs.

4.2 Proposed Solution

The proposed solution is an advanced ServiceNow data-ingestion and transformation framework. Import Sets receive the source data, while Custom Transform Maps control how records are mapped and transformed. Coalesce keys are used for duplicate handling; onBefore validation logic can validate incoming values; reference resolution supports related fields; and ACLs provide role-based access control.

4.3 Solution Architecture

1. SOURCE	2. IMPORT SET	3. TRANSFORM MAP	4. TARGET	5. CONTROL
CSV / Excel	Staging table Imported source records	Field mapping Coalesce Validation Reference resolution	ServiceNow target table	ACLs Logs Verification

The architecture separates source data from target records through the Import Set staging layer. The Transform Map acts as the transformation and mapping layer before data reaches the target table.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is organized into Agile sprints using Fibonacci-style story-point estimation. The planning structure uses Sprints, Epics, User Stories and Story Points, with velocity calculated as completed story points divided by the number of sprints.

Sprint	Epic	User Story / Task	SP
1	Data Import	Preparing Source Data (USN1)	2
1	Data Import	Creating Import Set (USN2)	2
1	Data Import	Loading Source Data (USN3)	1
1	Data Transformation	Creating Transform Map (USN4)	3
1	Data Transformation	Mapping Source and Target Fields (USN5)	3
1	Data Transformation	Coalesce / Duplicate Handling (USN6)	3
2	Data Validation	Validating Imported Records (USN7)	2
2	Data Validation	Handling Missing / Invalid Data (USN8)	3
2	Data Validation	Testing Reference Fields (USN9)	3
2	Deployment & Verification	Running Transform (USN10)	3
2	Deployment & Verification	Verifying Target Records (USN11)	2
2	Deployment & Verification	Error Handling & Final Testing (USN12)	3

Sprint 1 = 14 SP; Sprint 2 = 16 SP; Total = 30 SP; Number of Sprints = 2; Planned Velocity = 15 Story Points per Sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing covers the import workflow, transformation logic, duplicate handling, validation, reference fields, target verification, exception handling, and access control.

Test Area	Cases	Pass	Fail	Not Tested
Source Data / Import Set	5	5	0	0
Transform Map Configuration	5	5	0	0
Field Mapping & Coalesce	4	4	0	0
Data Validation	4	3	1	0
Reference Field Handling	3	3	0	0
Target Record Verification	4	4	0	0
Error / Exception Handling	3	3	0	0
Security / Access Control	2	2	0	0
TOTAL	30	29	1	0

UAT Summary: 30 test cases were included, with 29 passed, 1 failed, and 0 not tested. The failed validation case indicates that source-data validation requires correction before final acceptance.

7. RESULTS

The project establishes a structured workflow for importing source data into ServiceNow. The expected result is that source records are staged through an Import Set, transformed through the configured Transform Map, matched using Coalesce where required, validated, and loaded into the target table. The UAT summary records 29 passed cases and 1 failed validation case.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Bulk data can be loaded into ServiceNow.	Large batches can be affected by system performance limits.
Source fields can be mapped systematically.	Incorrect mappings can produce incorrect target data.
Coalesce can help prevent duplicate records.	Coalesce depends on an appropriate matching key.
Validation logic can reduce inconsistent data.	Validation rules require maintenance when source formats change.
Reference fields can be resolved during transformation.	Broken or unmatched references require investigation.
ACLs support secure role-based access.	Configuration and testing require administrator effort.

9. CONCLUSION

The Import Data using Transform Maps project provides a structured ServiceNow approach for bulk data onboarding. The project identifies problems around data quality, duplicates, mapping, references, and manual effort. The proposed Import Set and Transform Map workflow addresses these areas through controlled staging, transformation, Coalesce-based matching, validation, reference resolution, and role-based access control. The project is planned across two Agile sprints with a calculated velocity of 15 Story Points per Sprint.

10. FUTURE SCOPE

- Expand validation rules for more source-data formats.
- Introduce more automated reference-field resolution.

- Improve monitoring and reporting for large data batches.
- Add reusable Transform Map templates for different organizational datasets.
- Enhance duplicate-detection logic for complex datasets.
- Extend automated testing for future Transform Map changes.